

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

show ip route ospf

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? S0/0/0

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? 129

Does R2 show up as an OSPF neighbor on R1? Yes

Does R2 show up as an OSPF neighbor on R3? No

What does this information tell you?

Traffic to 192.168.2.0/24 from R3 will be passed through R1.

OSPF is not advertised across S0/0/1.

Traffic of R3 to 192.168.2.0/24 pass through 2 T1 links and R2 Gigabit 0/0 LAN link (i.e. 129 cost).

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

R2(config)

R2(config-router)

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? S0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

65. This is calculated by 1 T1 and R2 Gigabit 0/0 LAN link

Is R2 listed as an OSPF neighbor to R3? Yes

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

```
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

In order to obtain a more accurate cost for those links.

Higher default reference-bandwidth can support faster links.



Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

Cost to 192.168.3.0/24 from R1: $781 + 1 = 782$

Cost to 192.168.23.0/30 from R1: $781 + 64 = 845$



- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

1562. Route to 192.168.23.0/24 passes through two serial links, i.e. $781 + 781 = 1562$.



Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

OSPF choose the least accumulated cost route.

Cost through R2: $781 + 781 + 1 = 1563$

Cost through R1: $1565 + 1 = 1566$



Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

Router ID assignment control Designated and Backup Designated Router election process.

If the interface went down, the router ID that is associated with active interface can change.

Thus, it should be set with router-id command.

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2. Why is the DR/BDR election process not a concern in this lab?

The election process only occurs on multi-access network, while the links in this lab are point-to-point linked.

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3. Why would you want to set an OSPF interface to passive?

Configure interface to passive can eliminate unnecessary routing information, releasing bandwidth while still advertising network neighbours.

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4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

The hierarchical design of OSPF is achieved through its use of areas, allowing for efficient routing and minimized overhead, which supports the scalability of large networks.

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5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

router ospf [process-id] \n network 10.0.0.0 255.255.255.0 area 0

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6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

~~The OSPF route will be used for the routing decision since it has the lowest administrative distance of 110 compared to RIP's AD of 120 and EIGRP's AD of 90.~~

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